

Progress report Suishōdama

Smart India Hackathon 2026 · Internal Hackathon · Problem statement SIH26189

Problem statement

AI-Powered Criminal Network Analysis System — entity extraction, relationship mapping and key-influencer identification

Abstract

An investigating officer already holds the connections he needs. They sit unread across FIRs, call-detail records, bank statements, surveillance logs, criminal histories and intelligence reports, and finding them means holding hundreds of identifiers in his head at once. Suishōdama gives every case an investigator of its own.

Opening a case creates a self-contained file on disk carrying that case's knowledge graph, analytics, memory and hash-chained chain of custody. Documents are parsed deterministically, with no model anywhere in the ingest path, into people, phone numbers, vehicles, accounts, locations and organisations, each traceable to the character offset it was read from. The analytics layer computes the findings: person-projected centrality that treats a man and his SIM as one actor, Louvain community detection, and hidden brokers ranked person against person.

The officer types plain English. The assistant begins every question knowing nothing and queries the case brain through tools, so the case lives on disk rather than in a context window. It grows where a window only fills, it outlives the conversation, and no case can reach another. Every citation is verified against the graph before the officer sees it.

Work done till 10:00 PM on 8th September

Prototype complete and running end to end. All seven source kinds ingest deterministically; the demo case holds 125 nodes, 1,148 links and 61 custody entries. Graph, analytics, tool-using assistant, graph view and tamper-evident custody log all work. 38 tests pass; ingest is linear, 106 KB in 3.8 seconds.

Work left

Recording the demo video and preparing the final presentation for offline evaluation. On the system itself: rehearsing the six-minute demo against the running prototype, and widening evaluation beyond the trafficking case. OCR for scanned documents stays deliberately out of scope, so evidence is never read through a lossy step.

Challenges faced

1. **Ingest was too slow to survive a real case file.** 106 KB of documents took 63 seconds, and the cost was growing with the square of the input — so a full chargesheet bundle would never have finished at all. The ingest path was rebuilt to scale linearly. The same 106 KB now takes 3.8 seconds, and eight times the document costs 7.4 times the time instead of sixty-four.

2. **The most important man in the network scored near zero.** The problem statement asks the system to identify key influencers, and the measure for that is how many routes between people run through a person. But a kingpin whose only recorded link is to his own phone sits at a dead end — every route stops at the handset — so he ranked near the bottom. A man and his SIM are not two actors. Every phone, account, device and vehicle with a known holder is now folded into that holder before anything is scored, while identifiers nobody has attributed stay as themselves. That single choice is the difference between finding him and not.